B.Sc. (Ag.) I-Semester Examination, 2024 · Paper 23151-AG (AG-109)

Agricultural Heritage

(कृषि विरासत)

AGR-102 · Credit 1 (1+0)

Model Answer Paper (PYQ 2024 - all questions solved)

आदर्श उत्तर पत्र - पिछले वर्ष (2024) के सभी प्रश्न हल सहित

Time	Maximum Marks	Sections
3 Hours	50	A: 10 × 1 = 10 · B: 5 × 5 = 25 · C: 2 × 7.5 = 15

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SECTION A

Very Short Answer Type Questions

अति लघु उत्तरीय प्रश्न

Attempt all parts of Q.1. Answer each part in about 50 words.

प्रश्न 1 के सभी भागों के उत्तर दीजिए; प्रत्येक लगभग 50 शब्दों में।

10 × 1 = 10 Marks

Q.1 (i) Define shifting cultivation. — 1 Mark

स्थानान्तरित / झूम खेती को परिभाषित कीजिए।

Definition (परिभाषा)

A primitive system in which a forest patch is cleared by slash and burn, cropped for 2 - 3 years till fertility falls, then abandoned and the farmer moves to a new patch.

वन का टुकड़ा काटकर व जलाकर 2 - 3 वर्ष खेती, उर्वरता घटने पर उसे छोड़कर नई जगह खेती।

Local name	Region
Jhum	North-East India
Podu	Odisha, Andhra
Bewar / Dahiya	Madhya Pradesh
Kumari	Western Ghats

Q.1 (ii) What do you mean by GIAHS? — 1 Mark

जी.आई.ए.एच.एस. से आप क्या समझते हैं?

GIAHS (विश्व महत्त्व की कृषि विरासत प्रणाली)

Globally Important Agricultural Heritage Systems - outstanding traditional farming landscapes that conserve agro-biodiversity, livelihoods and indigenous knowledge; launched by FAO in 2002.

FAO (2002) द्वारा मान्य परम्परागत कृषि प्रणालियाँ जो जैव-विविधता, आजीविका व पारम्परिक ज्ञान संरक्षित करती हैं।

Indian GIAHS sites (भारत के स्थल)

- Saffron heritage, Pampore (J&K) - 2011
  केसर, पम्पोर
- Koraput tribal farming (Odisha) - 2012
  कोरापुट
- Kuttanad below-sea-level farming (Kerala) - 2013
  कुट्टनाड

Q.1 (iii) **Write down the names of two crops introduced by the Portuguese in India.** **1 Mark**

पुर्तगालियों द्वारा भारत में लाई गई दो फसलों के नाम लिखिए।

Answer (उत्तर)

Chilli and potato (also maize, tobacco, groundnut, cashew, pineapple, papaya, guava, sweet potato). Most came from the New World (Americas) in the 16th - 17th century.

मिर्च व आलू (साथ ही मक्का, तम्बाकू, मूंगफली, काजू, अनानास, पपीता, अमरूद)। ये अमेरिका से 16वीं - 17वीं सदी में आई।

Crop	Origin
Chilli	Mexico / S. America
Potato	Andes (Peru)
Cashew	Brazil
Tobacco	Americas

Q.1 (iv) **I.C.A.R.** **1 Mark**

आई.सी.ए.आर.

Indian Council of Agricultural Research (भारतीय कृषि अनुसंधान परिषद)

- Apex body for agricultural research and education in India; set up on 16 July 1929 (as Imperial Council) on the Royal Commission on Agriculture report.
  भारत में कृषि अनुसंधान व शिक्षा की शीर्ष संस्था; 16 जुलाई 1929 को स्थापित।
- Headquarters: Krishi Bhawan, New Delhi; works under DARE, Ministry of Agriculture and Farmers Welfare; President = Union Agriculture Minister.
  मुख्यालय कृषि भवन, नई दिल्ली; DARE के अधीन; अध्यक्ष केन्द्रीय कृषि मंत्री।

Q.1 (v) Uses of cow dung. — 1 Mark

गाय के गोबर का उपयोग।

Use	How	हिन्दी
Manure	FYM, compost, vermicompost - improves soil organic carbon	खाद
Biogas (gobar gas)	Anaerobic digestion gives methane (55 - 65%) fuel + slurry manure	बायोगैस
Natural farming inputs	Jeevamrut, Beejamrut, Panchagavya	जीवामृत, पंचगव्य
Fuel and plastering	Dung cakes (upla); mud-dung plaster of floors and walls	उपले, लिपाई

Q.1 (vi) Medicinal plants. — 1 Mark

औषधीय पौधे।

Meaning (अर्थ)

Plants whose parts contain active compounds used to prevent or cure disease; base of Ayurveda, Siddha and Unani systems.

जिन पौधों के भागों में रोग निवारक सक्रिय यौगिक होते हैं; आयुर्वेद का आधार।

Plant	Use / compound
Sarpagandha	Reserpine - blood pressure
Ashwagandha	Withanolides - tonic
Tulsi	Cough, immunity
Isabgol	Husk - constipation

Q.1 (vii) **Krishi-Panchang.** 1 Mark

कृषि-पंचांग।

Krishi-Panchang (कृषि-पंचांग)

A traditional agricultural almanac / calendar based on lunar months, tithi and nakshatra that tells farmers the right time for ploughing, sowing, transplanting, harvesting and rain forecast. Its roots are in texts like Krishi-Parashara and the folk sayings of Ghagh - Bhaddari.

चन्द्र मास, तिथि व नक्षत्र पर आधारित परम्परागत कृषि कैलेण्डर, जो जुताई, बुवाई, कटाई व वर्षा का समय बताता है; आधार: कृषि-पराशर, घाघ - भड्डरी की कहावतें।

Q.1 (viii) **Crop rotation.** 1 Mark

फसल चक्र।

Definition (परिभाषा)

Growing different crops in a planned sequence on the same field over years, so that soil fertility is kept and pests, diseases and weeds are reduced.

एक ही खेत में वर्षों तक योजनाबद्ध क्रम में अलग-अलग फसलें उगाना, जिससे उर्वरता बनी रहे व कीट-रोग घटें।

Rotation	Type
Rice - Wheat	1 year (100%)
Maize - Potato - Wheat	1 year (300%)
Cotton - Wheat - Moong	Includes legume

Q.1 (ix) Cash crop. — 1 Mark

नकदी फसल।

Definition (परिभाषा)

A crop grown mainly for sale in the market to earn money rather than for home consumption.

मुख्य रूप से बाज़ार में बेचकर धन कमाने हेतु उगाई जाने वाली फसल।

Examples (उदाहरण)

Sugarcane, cotton, jute, tobacco, tea, coffee, groundnut, mustard

गन्ना, कपास, जूट, तम्बाकू, चाय, कॉफी

Q.1 (x) Green revolution. — 1 Mark

हरित क्रान्ति।

Meaning (अर्थ)

Sharp rise in food grain (mainly wheat and rice) production from 1966 - 67 by using HYV seeds, fertilisers, irrigation and pesticides. Term coined by William Gaud (1968).

1966 - 67 से HYV बीज, उर्वरक, सिंचाई व कीटनाशी से गेहूँ - धान उत्पादन में तेज़ वृद्धि; शब्द विलियम गॉड (1968)।

Person / input	Role
Norman Borlaug	Father (world); dwarf wheat
M.S. Swaminathan	Father in India
Lerma Rojo 64A, Sonora 64	Mexican wheat
IR 8	Miracle rice

SECTION B

Short Answer Type Questions

लघु उत्तरीय प्रश्न

Attempt any five questions. Answer each in about 200 words. (All eight solved here)

किन्हीं पाँच प्रश्नों के उत्तर दीजिए; प्रत्येक लगभग 200 शब्द। (यहाँ सभी आठ हल)

5 × 5 = 25 Marks

Q.2 Differentiate between mixed farming and specialised farming. 5 Marks

मिश्रित खेती एवं विशिष्ट खेती में अन्तर स्पष्ट कीजिए।

Mixed farming: crop production combined with livestock rearing on the same farm, each supporting the other. Specialised farming: the farm depends on one enterprise or crop (60% or more income) for the market.

मिश्रित खेती: फसल उत्पादन के साथ पशुपालन। विशिष्ट खेती: एक ही उद्यम / फसल से 60% या अधिक आय।

Basis	Mixed farming (मिश्रित खेती)	Specialised farming (विशिष्ट खेती)
Enterprises	Crops + dairy / poultry / goat	Single crop or enterprise, e.g. tea, sugarcane, dairy
Risk	Low - loss in one is covered by other	High - price or weather failure hits whole income
Use of resources	By-products recycled (dung → manure, straw → fodder)	Little recycling; purchased inputs
Income flow	Regular round the year (milk, eggs)	Seasonal, at harvest
Skill and capital	Moderate, family labour	High technical skill, more capital, machinery
Suitable for	Small and marginal farmers	Large farms with assured market
Example	Wheat - rice + 2 buffaloes in UP	Tea estates of Assam, grape farms of Nashik

Q.3 What are the criteria for recognition of GIAHS? — 5 Marks

जी.आई.ए.एच.एस. पहचान के लिए कौन-कौन सी कसौटियाँ / शर्तें हैं?

FAO selects a GIAHS on the basis of five criteria, plus an action plan for dynamic conservation prepared with local people.

FAO पाँच कसौटियों तथा स्थानीय लोगों के साथ बनी संरक्षण कार्य-योजना के आधार पर GIAHS चुनता है।

1. Food and livelihood — खाद्य व आजीविका सुरक्षा

2. Agro-biodiversity — कृषि जैव-विविधता

3. Local knowledge — पारम्परिक ज्ञान प्रणाली

4. Culture — संस्कृति व सामाजिक संगठन

5. Landscape — विशिष्ट भू-दृश्य

Criterion	What FAO looks for
Food and livelihood security	System feeds and supports the local community
Agro-biodiversity	Rich diversity of crops, varieties, animals, wild relatives
Traditional knowledge	Indigenous practices of soil, water, pest and crop management
Culture and social organisation	Festivals, beliefs, community rules linked to farming
Landscapes / seascapes	Unique land and water use pattern, e.g. Kuttanad below sea level

Q.4 Write down the features of agriculture during the Vedic period.

5 Marks

वैदिक काल में भारत में कृषि का स्वरूप कैसा था, लिखिए।

Rigvedic period (1500 - 1000 BC) (ऋग्वैदिक काल)

- Pastoral + farming; cattle = wealth (*gau*)
  पशुपालन प्रधान; गाय धन
- Wooden plough langala / sira drawn by oxen
  लकड़ी का हल - लांगल / सीर
- Main crop yava (barley); later rice (vrihi)
  मुख्य फसल यव (जौ)
- Irrigation from wells (*avata*) and channels (*kulya*)
  कुआँ (अवट), नहर (कुल्या)

Later Vedic period (1000 - 600 BC) (उत्तर वैदिक काल)

- Agriculture becomes main occupation; iron tools
  कृषि मुख्य व्यवसाय; लोहे के औज़ार
- Rice, wheat, pulses, sesame grown
  धान, गेहूँ, दालें, तिल
- Land types: urvara (fertile), usara (barren)
  उर्वरा व ऊसर भूमि
- Manure (*karisha* - cow dung) used
  करीष (गोबर) खाद

Other features (अन्य विशेषताएँ)

Atharvaveda has prayers for rain and charms against pests; Krishi was respected and the plough was worshipped; six to twelve oxen pulled heavy ploughs; two crops a year were known.

अथर्ववेद में वर्षा की प्रार्थना व कीट निवारण मन्त्र; हल की पूजा; वर्ष में दो फसलें।

Q.5 What were the ancient methods of water harvesting in India?

5 Marks

भारत में जल संचयन की प्राचीन विधियाँ कौन सी थीं?

Indians built region-specific structures to capture rain and stream water; many still work today.

भारत में क्षेत्र के अनुसार वर्षा व धारा का जल रोकने की संरचनाएँ बनीं, जिनमें से कई आज भी उपयोगी हैं।

Structure	Region	Feature
Kallanai (Grand Anicut)	Tamil Nadu (Kaveri)	Stone dam by Karikala Chola, about 2nd century AD
Sudarshan lake	Girnar, Gujarat	Reservoir of Chandragupta Maurya's time; repaired by Rudradaman
Eri (tanks)	Tamil Nadu	Chain of village tanks for paddy
Ahar - Pyne	South Bihar	Catchment basin (ahar) fed by channels (pyne)
Khadin, Johad, Kund / Tanka	Rajasthan	Earthen bunds, ponds, covered underground tanks
Kuhl	Himachal, J&K	Gravity channels diverting glacier streams
Zabo	Nagaland	Forest - pond - paddy - livestock system
Bamboo drip	Meghalaya	Bamboo pipes carry spring water to betel vine
Baoli / step well	Gujarat, Rajasthan, Delhi	Stepped wells reaching groundwater

Q.6 What do you mean by Indigenous Technical Knowledge? 5 Marks

स्वदेशी तकनीकी ज्ञान से आप क्या समझते हैं?

Definition (परिभाषा)

Indigenous Technical Knowledge (ITK) is the knowledge of farming developed by local communities through experience over generations, passed on orally, low-cost, eco-friendly and suited to local conditions.

स्थानीय समुदायों द्वारा पीढ़ियों के अनुभव से विकसित, मौखिक रूप से हस्तांतरित, सस्ता, पर्यावरण-हितैषी कृषि ज्ञान।

Area	ITK practice	Scientific basis
Grain storage	Mixing dry neem leaves / ash with stored grain	Azadirachtin repels insects; ash blocks movement
Pest control	Spray of cow urine, neem or garlic - chilli extract	Repellent and antifeedant action
Soil fertility	Sheep / goat penning in fields; green leaf manure	Adds dung, urine, nitrogen
Weather forecast	Ghagh - Bhaddari sayings, ant movement, cloud colour	Long observation of nature
Seed treatment	Seeds coated with cow dung slurry before sowing	Beneficial microbes, protection

Value: sustainable, cheap, based on local resources; can be validated and blended with modern science.

महत्त्व: टिकाऊ, सस्ता; आधुनिक विज्ञान के साथ जोड़ा जा सकता है।

Q.7 Explain crop voyage in India.

5 Marks

भारत में फसल यात्रा का वर्णन कीजिए।

Crop voyage is the movement of crops from their centre of origin to new lands through traders, invaders, travellers and colonial rulers. India both received and gave many crops.

फसल यात्रा = फसलों का उद्गम स्थान से व्यापारियों, आक्रमणकारियों, यात्रियों द्वारा नए क्षेत्रों में पहुँचना। भारत ने फसलें पाईं भी और दीं भी।

Crops that came to India (भारत आई फसलें)

Crop	From / through
Maize, potato, chilli, tomato, tobacco, groundnut, cashew, papaya	Americas, by Portuguese
Coffee	Yemen, by Baba Budan (17th century)
Sorghum, pearl millet, finger millet	Africa (ancient)
Chinese tea, cinchona	British period

Crops India gave the world (भारत से गई फसलें)

Crop	Note
Rice, sugarcane	Indian / Indo-Burma centre
Cotton (G. arboreum)	Mohenjo-daro cloth
Black pepper, cardamom	Spice trade to Rome, Arabs
Mango, banana, brinjal, cucumber	Native fruits and vegetables

Importance: widened food basket; many introduced crops (potato, chilli, maize) are now staples of India.

महत्त्व: भोजन विविधता बढ़ी; आलू, मिर्च, मक्का आज भारत की मुख्य फसलें हैं।

Q.8 What are the agricultural resources? Describe in brief.

5 Marks

कृषि संसाधन कौन से हैं? संक्षिप्त वर्णन करिए।

Agricultural resources are the natural, biological and human inputs on which farm production depends.

कृषि संसाधन = वे प्राकृतिक, जैविक व मानवीय साधन जिन पर कृषि उत्पादन निर्भर है।

Resource	India's position	हिन्दी
Land	Geographical area 328.7 Mha; about 140 Mha net sown area (largest arable land after USA)	भूमि
Soil	Alluvial (largest), black, red, laterite, desert, mountain soils	मृदा
Water	Average rainfall about 1,170 mm; rivers, groundwater; about half area irrigated	जल
Climate	Tropical to temperate; 15 agro-climatic regions allow crops round the year	जलवायु
Livestock	Largest livestock population (535.8 million, 20th Census 2019)	पशुधन
Biodiversity	One of 17 mega-diverse countries; Indo-Burma centre of origin	जैव-विविधता
Human resource	Nearly half of the workforce engaged in agriculture	मानव संसाधन
Institutions	ICAR, SAUs, KVKs, cooperatives, NABARD	संस्थागत संसाधन

Q.9 **Write the full form of: (a) IARI (b) IIVR (c) NABARD (d) NDDB (e) KVK** **5 Marks**

निम्नलिखित का पूरा नाम लिखिए: (अ) IARI (ब) IIVR (स) NABARD (द) NDDB (य) KVK

Abbreviation	Full form	Key fact
IARI	Indian Agricultural Research Institute भारतीय कृषि अनुसंधान संस्थान	Pusa, New Delhi; founded 1905 at Pusa (Bihar)
IIVR	Indian Institute of Vegetable Research भारतीय सब्जी अनुसंधान संस्थान	Varanasi, Uttar Pradesh
NABARD	National Bank for Agriculture and Rural Development राष्ट्रीय कृषि एवं ग्रामीण विकास बैंक	Set up 12 July 1982; HQ Mumbai
NDDB	National Dairy Development Board राष्ट्रीय डेयरी विकास बोर्ड	1965, Anand (Gujarat); Operation Flood
KVK	Krishi Vigyan Kendra कृषि विज्ञान केन्द्र	District farm science centre; first at Puducherry (1974)

SECTION C

Long Answer Type Questions

दीर्घ उत्तरीय प्रश्न

Attempt any two questions. Answer each in about 500 words. (All four solved here)

किन्हीं दो प्रश्नों के उत्तर दीजिए; प्रत्येक लगभग 500 शब्द। (यहाँ सभी चार हल)

2 × 7.5 = 15 Marks

Q.10 **Define agricultural heritage. Explain the agricultural heritage recognised in India.** **7.5 Marks**

कृषि विरासत को परिभाषित कीजिए। भारत में पहचान किये गये कृषि विरासत का वर्णन कीजिए।

Definition (परिभाषा)

Agricultural heritage is the inherited wealth of farming - traditional practices, tools, crop varieties, animal breeds, water systems, texts, festivals and landscapes - developed by our ancestors over thousands of years and still useful for sustainable agriculture.

हज़ारों वर्षों में पूर्वजों द्वारा विकसित कृषि पद्धतियाँ, औज़ार, किस्में, नस्लें, जल प्रणालियाँ, ग्रन्थ, त्योहार व भू-दृश्य - जो आज भी टिकाऊ खेती में उपयोगी हैं।

Component	Examples from India
Ancient texts	Arthashastra (Kautilya), Brihat Samhita (Varahamihira), Krishi-Parashara, Vrikshayurveda (Surapala), Kashyapiya Krishisukti
Traditional practices	Mixed cropping, crop rotation, green manuring, ITK pest control
Water heritage	Kallanai, Eri tanks, Ahar - Pyne, Kuhl, Zabo, step wells
Genetic heritage	Kalanamak rice, Desi cotton, Gir and Sahiwal cows, native millets
Festivals and culture	Pongal, Baisakhi, Onam, Makar Sankranti, Hal Shashthi

Q.10 GIAHS sites of India (recognised by FAO)

contd.

भारत के GIAHS स्थल

Saffron heritage (केसर विरासत, पम्पोर)

- Pampore, Kashmir - 2011
  पम्पोर, कश्मीर
- Karewa soils; saffron with pulses, almond
  करेवा मृदा
- Hand picking and drying knowledge
  पारम्परिक तुड़ाई

Koraput (कोरापुट, ओडिशा)

- Tribal farming, Odisha - 2012
  जनजातीय कृषि
- Hundreds of native rice landraces, millets
  धान की देसी किस्में
- Considered a centre of origin of rice
  धान का उद्गम क्षेत्र

Kuttanad (कुट्टनाड, केरल)

- Below-sea-level farming - 2013
  समुद्र तल से नीचे खेती
- Rice - fish in polders (dykes)
  धान - मछली
- Salt water controlled by Thanneermukkom bund
  खारे पानी पर नियन्त्रण

Other traditional systems valued in India (अन्य परम्परागत प्रणालियाँ)

System	Place	Feature
Apatani rice - fish	Ziro, Arunachal Pradesh	Wet rice with fish, no animal or tractor power
Zabo	Nagaland	Forest - water pond - paddy - livestock linkage
Bamboo drip	Meghalaya	Gravity water through bamboo pipes

Conclusion: conserving such heritage keeps biodiversity, culture and climate-resilient farming alive.

निष्कर्ष: इस विरासत का संरक्षण जैव-विविधता व जलवायु-सहिष्णु खेती के लिए आवश्यक है।

Q.11 Elaborate the economic / agronomic classification of crops. 7.5 Marks

फसलों का आर्थिक / शस्यकी वर्गीकरण विस्तार से कीजिए।

Crops are grouped by their economic use (agronomic classification) so that their cultivation, marketing and research can be planned.

फसलों को उनके आर्थिक उपयोग के आधार पर वर्गीकृत किया जाता है ताकि खेती, विपणन व अनुसंधान की योजना बने।

Group	Use	Examples
Cereals (अनाज)	Grain, staple food; family Poaceae	Rice, wheat, maize, barley
Millets (मोटे अनाज)	Small-grained, hardy cereals	Sorghum, pearl millet, finger millet, kodo
Pulses (दालें)	Protein-rich seeds; fix nitrogen	Chickpea, pigeonpea, lentil, moong, urd
Oilseeds (तिलहन)	Edible / industrial oil	Mustard, groundnut, soybean, sesame, sunflower
Fibre crops (रेशा)	Fibre for cloth, ropes	Cotton, jute, mesta, sunnhemp
Sugar crops (शर्करा)	Sugar, jaggery	Sugarcane, sugarbeet
Starch / tuber crops	Starch, food	Potato, sweet potato, tapioca
Forage crops (चारा)	Green fodder, hay, silage	Berseem, oat, napier, sorghum
Beverage and narcotics	Drinks, stimulants	Tea, coffee, tobacco
Spices and condiments	Flavour	Chilli, turmeric, coriander, cumin

Q.11 Other important bases of crop classification

contd.

फसल वर्गीकरण के अन्य आधार

Based on season (ऋतु के आधार पर)

Season	Sown / harvested	Crops
Kharif	June - July / Sept - Oct	Rice, maize, cotton, arhar
Rabi	Oct - Nov / March - April	Wheat, gram, mustard
Zaid	March / June	Watermelon, moong, cucumber

Based on life span (जीवन अवधि)

Type	Example
Annual (one season)	Wheat, rice
Biennial (two seasons)	Sugarbeet, onion (seed)
Perennial (many years)	Napier, lucerne, sugarcane ratoon

Special purpose crops (विशेष उद्देश्य फसलें)

- Catch crop - short crop between two main crops (moong)
  बीच की अल्पावधि फसल
- Cover crop - protects soil from erosion (cowpea)
  आवरण फसल
- Green manure - ploughed in (dhaincha, sunnhemp)
  हरी खाद
- Trap crop - attracts pests (marigold in tomato)
  प्रपंच फसल

Based on root system / photoperiod (जड़ व दीप्तिकाल)

- Tap root (gram, mustard) vs fibrous root (wheat, rice)
  मूसला व झकड़ा जड़
- Short day (rice, soybean), long day (wheat), day neutral (maize)
  अल्प, दीर्घ व उदासीन दिवस

Q.12 Describe in detail the importance of agriculture in India. 7.5 Marks

भारत में कृषि के महत्त्व का विस्तार से वर्णन कीजिए।

Agriculture is called the backbone of the Indian economy; it feeds 1.4 billion people and gives livelihood to nearly half of the workforce.

कृषि भारतीय अर्थव्यवस्था की रीढ़ है; 140 करोड़ लोगों का भोजन व लगभग आधे कार्यबल की आजीविका।



Role	Explanation
Food security	Record food grain production (over 330 million tonnes) supports the PM Garib Kalyan Anna Yojana
Employment	About 46% of the workforce depends on agriculture and allied sectors
National income	Agriculture and allied sectors give about 18% of Gross Value Added

Q.12 Importance of agriculture (continued)

contd.

कृषि का महत्त्व (आगे)

Role	Explanation	हिन्दी
Raw material	Cotton - textile, sugarcane - sugar, jute - bags, oilseeds - oil mills	कच्चा माल
Exports and foreign exchange	Rice, spices, tea, coffee, marine products, sugar, fruits	निर्यात
Market for industry	Demand for tractors, fertilisers, pesticides, consumer goods in villages	उद्योगों का बाज़ार
Livestock and nutrition	India is the largest milk producer; dairy is daily cash income	दुग्ध उत्पादन
Capital formation	Rural savings and taxes support development	पूँजी निर्माण
Ecology and culture	Festivals, forests, soil and water conservation linked to farming	संस्कृति व पर्यावरण

Conclusion (निष्कर्ष)

Growth in agriculture reduces poverty faster than growth in any other sector; hence modern, sustainable and profitable farming is essential for a developed India (Viksit Bharat 2047).

कृषि विकास गरीबी को सबसे तेज़ घटाता है; अतः विकसित भारत के लिए आधुनिक व टिकाऊ कृषि आवश्यक है।

Q.13 Explain the current scenario of Indian agriculture.

7.5 Marks

भारतीय कृषि के वर्तमान परिदृश्य का वर्णन कीजिए।

Strengths today (वर्तमान उपलब्धियाँ)

- Largest producer of milk, pulses, jute; among top two in rice, wheat, sugarcane, fruits and vegetables
  दूध, दालें, जूट में प्रथम
- Food grain surplus; major exporter of rice
  खाद्यान्न अधिशेष; चावल निर्यातक
- Horticulture output now more than food grains
  बागवानी उत्पादन खाद्यान्न से अधिक
- Fast growth of dairy, poultry, fisheries
  पशुपालन व मत्स्य में तेज़ वृद्धि

Major schemes (प्रमुख योजनाएँ)

Scheme	Purpose
PM-KISAN (2019)	Rs 6,000 per year income support
PMFBY (2016)	Crop insurance
Soil Health Card (2015)	Soil test-based fertiliser use
PMKSY (2015)	Per drop more crop
e-NAM (2016)	Online mandi trading
10,000 FPO (2020)	Collective marketing

Technology trends: drones (Namo Drone Didi), digital agriculture mission, natural farming, micro-irrigation, biofortified varieties.

तकनीकी रुझान: ड्रोन, डिजिटल कृषि मिशन, प्राकृतिक खेती, सूक्ष्म सिंचाई, जैव-पोषित किस्में।

Q.13 Problems and future prospects of Indian agriculture

contd.

भारतीय कृषि की समस्याएँ व भविष्य

Problems (समस्याएँ)

- Small holdings: average about 1.08 ha; 86% small and marginal farmers
  छोटी जोत
- Falling groundwater; half area still rainfed
  भूजल गिरावट
- Soil health decline, imbalanced NPK use
  मृदा स्वास्थ्य में गिरावट
- Climate change: heat waves, erratic rain
  जलवायु परिवर्तन
- Post-harvest losses, weak market price
  कटाई उपरान्त हानि, कम मूल्य

Future prospects (भविष्य की सम्भावनाएँ)

- Crop diversification to pulses, oilseeds, horticulture
  फसल विविधीकरण
- Precision farming, drones, AI advisories
  परिशुद्ध खेती
- Natural and organic farming for export
  प्राकृतिक / जैविक खेती
- Food processing and value chains via FPOs
  खाद्य प्रसंस्करण
- Climate-resilient varieties and water saving
  जलवायु-सहिष्णु किस्में

Conclusion (निष्कर्ष)

Indian agriculture has moved from food shortage to surplus; the next goal is higher farmer income with sustainability.

भारतीय कृषि कमी से अधिशेष तक पहुँची; अगला लक्ष्य टिकाऊपन के साथ किसान की आय बढ़ाना।